

Quantum Private Distributed Matrix Multiplication: New Families of Quantum Codes

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Abstract—We consider the problem of private distributed matrix multiplication (PDMM) in the quantum setting (QPDMM). In the most general version of this problem, a user who has two large matrices A and B , which are partitioned into K and L sub-matrices, respectively, wants to compute their product without having enough computational resources. Thus, the user relays these sub-matrices to honest but curious N servers. Since the user needs to keep their data private, they have to hide them from any T colluding servers. Since the servers are honest, once the user receives the responses, the required multiplication must be decodable. In the quantum setting, we allow the servers to share an entangled state among each other and respond back to the user over separate quantum channels. In previous work, we have shown that the classical schemes, namely, gap additive secure polynomial (GASP), modified GASP, cyclic-addition degree tables (CAT) and discretely optimized GASP (DOG) can be extended to their quantum counterparts if a certain feasibility condition is satisfied. Furthermore, we showed analytically that if $T \geq KL - K + 1$, the GASP code can be extended to the quantum counterpart. Therefore, there is a need to design special quantum codes to solve the QPDMM problem when the feasibility condition is not satisfied.

In this paper, we consider the QPDMM problem when the feasibility condition fails. We design new codes that are used solely for the quantum setting to take advantage of the availability of quantum resources. We divide the problem into two separate regimes (assuming $K \geq L$, without loss in generality): the high privacy regime when $T \geq K \geq L$ or $K \geq T \geq L$, and the low privacy regime when $K \geq L > T$. The reason for separating the two regimes stems from the fact that each regime requires a different technique to design an efficient QPDMM code. The main performance metric for any PDMM or QPDMM code is the ratio between the dimensions of the required product and the number of servers needed, where the fewer the servers, the more efficient the code is. We compare the rates for the quantum codes that we designed to the state-of-the-art classical codes with the same parameters. In the high privacy regime, we show that super-dense coding gain is achieved in certain scenarios, i.e., we double the classical rate. In addition, for the low privacy regime, a gain of 1.5 is achievable in certain scenarios.

Index Terms—Non-feasibility to extend classical codes, outer product partitioning, private distributed matrix multiplication, quantum codes, super-dense coding gain.

I. INTRODUCTION

Matrix multiplication is a crucial step in many machine learning algorithms. Nowadays, most non-trivial applications require data with very high dimensionality in the training phase [1]–[3]. Edge devices or personal computers are still not equipped with enough computational capabilities to deal

with the fast-growing dimensionality. Thus, a solution proposed by the computer science community is to relay the computation privately to remote servers which have sufficient computational power [4]–[9].

Recently, the communication and information theory communities have adopted this computation-outsourcing model. In [10], the private distributed matrix multiplication (PDMM) problem was formulated in an information theoretic model, where the two matrices A and B must be kept information theoretically private from any T colluding servers, i.e., even if any T servers out of the total N computing servers share the transmitted information by the user they cannot get any information about the matrices A and B .

Three main variants were then adopted in the literature for the PDMM problem: outer product partitioning (OPP), inner product partitioning (IPP), and batch multiplication. In OPP, A is divided horizontally into K sub-matrices and B is divided vertically into L sub-matrices. In IPP, A is divided vertically into L sub-matrices and B is divided horizontally into K sub-matrices. In batch multiplication, A is divided horizontally into K sub-matrices and B is divided vertically into L sub-matrices. In IPP, A is divided vertically into L sub-matrices and B is divided horizontally into K sub-matrices. IPP is then represented by $\sum_{i=1}^K A_i B_i$ [11]–[13]. Finally, batch multiplication is to compute the multiplication between subblocks with the same indices only, i.e., $[A_1 B_1, A_2 B_2, \dots, A_K B_K]$. Each of the aforementioned settings has its own separate line of work in the classical setting, and the codes developed for each setting are not optimal for other settings.

In this paper, we focus on the first setting, i.e., OPP. It has applications in numerous fields such as tensor contraction, deep learning and GPU acceleration [14]–[16]. For a deep dive in the OPP literature, see for example, [17]–[21]. There are two main works in the literature that we focus on here: the work related to gap additive secure polynomial (GASP) codes [22], [23], and the work related to modified GASP codes which are cyclic-addition degree table (CAT) and discretely optimized GASP (DOG) codes [24]. The main reason to focus on these codes is their efficiency in terms of rates for the OPP problem, i.e., they provide the highest ratio between the dimensions of the required product per server computation. We use them to compare the rate we achieve in the quantum setting and the highest-rate code for the classical counterpart.

Concurrently, quantum communication has gained significant interest since the discovery of super-dense coding [25]. In a nutshell, it was shown that pre-entanglement between a transmitter and a receiver allows one qubit to carry two bits. This phenomenon is coined as super-dense coding gain. More recently, in a setting with N entangled transmitters, it was

shown that, in certain applications such as quantum private information retrieval (QPIR) [26]–[30], distributed storage [31], [32], and quantum computation [33], a gain similar to the super-dense coding gain is achievable under certain regimes. In these applications, the transmitters are entangled, unlike the main setting for super-dense coding where a transmitter and a receiver are entangled. Although it was shown that when N transmitters are entangled, the receiver can get only N bits, the super-dense gain is still achievable. In the context of PDMM specifically, we were able to show in previous work that the same holds for the QPDMM problem as well, provided a certain feasibility condition is satisfied in [34].

In this paper, we aim to look deeper into the QPDMM problem beyond the feasibility condition. We design quantum codes specifically for the QPDMM problem when the feasibility condition fails. We divide the problem into two separate regimes, the high privacy regime and the low privacy regime. In the high privacy regime, we design several codes for different scenarios, e.g., $K = L = T = n^2$ and $K = T = n^k$ with $L = n^\ell$. The designs are based on the generalized Reed Solomon (GRS) codes and dual GRS codes. In the low privacy setting, we adopt approaches from the QPIR problem, where the privacy parameter is increased slightly to make use of the quantum resources. For both regimes, we plot the resulting rates with the state-of-the-art classical code counterpart. The results show varying gains that peak at 1.8 for the high privacy regime with sporadic appearance of super-dense coding gains in very specific settings. In addition, in the low privacy regime, the results show gains that peak at 1.5.

Notation: Throughout the paper, we use the following notation, unless stated otherwise. We denote the natural numbers as $\mathbb{N} = \{0, 1, 2, \dots\}$. Let $\alpha \in \mathbb{N}^{K+T}$ and $\beta \in \mathbb{N}^{L+T}$, then we use $\alpha \oplus \beta$ to denote

$$\alpha \oplus \beta = \begin{bmatrix} \alpha_1 + \beta_1 & \alpha_1 + \beta_2 & \dots & \alpha_1 + \beta_{L+T} \\ \alpha_2 + \beta_1 & \alpha_2 + \beta_2 & \dots & \alpha_2 + \beta_{L+T} \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{K+T} + \beta_1 & \alpha_{K+T} + \beta_2 & \dots & \alpha_{K+T} + \beta_{L+T} \end{bmatrix}. \quad (1)$$

If $A \in \mathbb{R}^{m \times n}$, then let $\text{Set}(A)$ denote the set with the elements of the matrix A . We use the superscript t for the transpose operator, with computational basis as the main basis, and $\dagger$ is the conjugate transpose operator. $[n]$ denotes the set $\{1, \dots, n\}$, and $[a : b]$ denotes the set $\{a, a+1, \dots, b\}$. For a matrix $A \in \mathbb{R}^{m \times n}$, $A(:, [a : b])$ defines a sub-block of the aforementioned matrix with all the rows included and the columns chosen are those indexed by the set $[a : b]$. All vectors are column vectors except stated otherwise. All symbols that denote sets are written using a calligraphic font. If $\mathcal{S}$ is a set, then $\mathcal{S}(m)$ denotes the m th element in the set after a predefined indexing. If a is any constant, then $a^{\mathcal{S}} = [a^{\mathcal{S}(1)}, \dots, a^{\mathcal{S}(|\mathcal{S}|)}]$. If K is a constant and $a = [a_1, \dots, a_n]$, then, $K + a = K\mathbf{1}^t + a$, where $\mathbf{1}$ is the all ones vector with the required length. We use $\mathbb{1}_{\mathcal{S}}$ as an indicator random variable for the occurrence of the event $\mathcal{S}$. I_N is the identity matrix of size N . Uppercase letters are used for matrices, and lowercase letters are used for vectors. Constants are written in

upper or lower case fonts depending on the situation, and will be mentioned if not clear from the context. Finally, for the quantum systems and quantum operators, we use uppercase with typewriter font, and for the associated quantum density matrix, as a special case of quantum operators, we use the Greek symbols σ and ρ .

II. PRELIMINARIES

A. Dual Codes

Let C be a linear code in $\mathbb{F}_q^n$, the dual linear code of C , $C^\perp$, is defined as

$$C^\perp = \{x \in \mathbb{F}_q^n : x^t c = 0, c \in C\}. \quad (2)$$

Since both are linear codes, they can be generated using generator matrices, $G_C \in \mathbb{F}_q^{k \times n}$, and $G_{C^\perp} \in \mathbb{F}_q^{n-k \times n}$, respectively. Then, they are connected with the relation

$$G_C G_{C^\perp}^t = 0. \quad (3)$$

An important code family in this work is the GRS code and its dual [35]. The $[n, k, d]_q$ GRS code is defined as follows.

Definition 1 (GRS code) Let $F = \mathbb{F}_q$ be a field and choose n distinct non-zero elements $\alpha_1, \dots, \alpha_n$. Let $u_1, \dots, u_n$ be non-zero elements in the field. Then, for any $0 \leq k \leq n$, the GRS is defined as

$$\text{GRS}(\alpha, u) = \{(u_1 f(\alpha_1), u_2 f(\alpha_2), \dots, u_n f(\alpha_n)) : f(x) \in F[x]_k\}, \quad (4)$$

where $\alpha = \alpha_{[n]}$, $u = u_{[n]}$, and $F[x]_k$ is the polynomial ring with degree less than k . In addition, the corresponding canonical generator matrix (transposed) is given by

$$G_{\text{GRS}(\alpha, u)} = \begin{bmatrix} u_1 & u_1 \alpha_1 & u_1 \alpha_1^2 & \dots & u_1 \alpha_1^{k-1} \\ u_2 & u_2 \alpha_2 & u_2 \alpha_2^2 & \dots & u_2 \alpha_2^{k-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ u_n & u_n \alpha_n & u_n \alpha_n^2 & \dots & u_n \alpha_n^{k-1} \end{bmatrix}. \quad (5)$$

For notational convenience, we henceforth refer to the transposed generator matrix simply as the generator matrix without loss of generality.

Theorem 1 (Dual GRS) Given an $[n, k, d_1]_q$ GRS code parameters with α and u as the generating elements, the dual code is also a $[n, n-k, d_2]_q$ GRS code with α and $v = v_{[n]}$ as the generating elements, where

$$v_i = \frac{1}{u_i} \left(\prod_{\substack{j \in [n] \\ j \neq i}} (\alpha_j - \alpha_i) \right)^{-1}. \quad (6)$$

From Theorem 1, the generator matrix for the dual GRS can be written as

$$G_{\text{GRS}(\alpha, v)} = \begin{bmatrix} v_1 & v_1 \alpha_1 & v_1 \alpha_1^2 & \dots & v_1 \alpha_1^{n-k-1} \\ v_2 & v_2 \alpha_2 & v_2 \alpha_2^2 & \dots & v_2 \alpha_2^{n-k-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ v_n & v_n \alpha_n & v_n \alpha_n^2 & \dots & v_n \alpha_n^{n-k-1} \end{bmatrix}. \quad (7)$$

B. Dual of the Shifted GRS Code

Definition 2 (Shifted GRS) An $[n, k, d]_q$ GRS code is called ℓ -shifted GRS if the generator matrix is defined as

$$G_{GRS(\alpha, u)} = \begin{bmatrix} u_1 \alpha_1^\ell & u_1 \alpha_1^{\ell+1} & u_1 \alpha_1^{\ell+2} & \dots & u_1 \alpha_1^{\ell+k-1} \\ u_2 \alpha_2^\ell & u_2 \alpha_2^{\ell+1} & u_2 \alpha_2^{\ell+2} & \dots & u_2 \alpha_2^{\ell+k-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ u_n \alpha_n^\ell & u_n \alpha_n^{\ell+1} & u_n \alpha_n^{\ell+2} & \dots & u_n \alpha_n^{\ell+k-1} \end{bmatrix}. \quad (8)$$

Theorem 2 For any $[n, k, d_1]_q$ ℓ_1 -shifted GRS code with u and α as the generating elements, there exists an $[n, n-k, d_2]_q$ ℓ_2 -shifted GRS code with v and α as its dual code if

$$v_i = \frac{1}{u_i \alpha_i^{\ell_1 + \ell_2}} \left(\prod_{\substack{j \in [n] \\ j \neq i}} (\alpha_j - \alpha_i) \right)^{-1}. \quad (9)$$

This result follows directly by substituting (6) and plugging u_i as $u_i \alpha_i^{\ell_1}$ and v_i as $v_i \alpha_i^{\ell_2}$, respectively.

C. Quantum Information Theory

For completeness, we recall the following standard definitions from quantum information theory. Detailed explanation and properties can be found in standard references on quantum information processing, e.g., [36].

Definition 3 (Quantum density matrices) For a quantum system A that can be in the state $|\psi_j\rangle$ with probability p_j , the quantum density matrix ρ_A is defined as

$$\rho_A = \sum_j p_j |\psi_j\rangle \langle \psi_j|, \quad (10)$$

with $p_j \geq 0$ and $\sum_j p_j = 1$.

Definition 4 (Quantum operation) A quantum operation E from the quantum system A to the quantum system B is a linear, completely-positive, and trace-preserving map from the set of all density operators of the Hilbert space representing A to the density operators of the Hilbert space representing B .

Definition 5 (Trace and partial trace) Given a quantum operator X in a Hilbert space with dimension d , the trace operator is defined as

$$\text{tr}(X) = \sum_{i=0}^{d-1} \langle i | X | i \rangle. \quad (11)$$

In addition, given a composite quantum state ρ_{AB} , the partial trace for the quantum system A is given by

$$\rho_B = \text{tr}_A(\rho_{AB}) = \sum_{i=0}^{d_A-1} (\langle i | \otimes I_{d_B}) \rho_{AB} (|i\rangle \otimes I_{d_B}), \quad (12)$$

where d_A and d_B are the dimensions of the Hilbert spaces associated with the quantum systems A and B , respectively.

Theorem 3 (Kraus representation) Any quantum operation E acting on a quantum state ρ can be written in the form

$$E(\rho) = \sum_i M_i \rho M_i^\dagger, \quad (13)$$

where $\sum_i M_i^\dagger M_i = I$.

D. N -Sum Box Abstraction

The N -sum box abstraction describes an approach for encoding the classical dits onto quantum states when the transmitters are entangled [37]. The encoding and decoding structure in the N -sum box abstraction is described as follows. In the encoding stage, the servers use generalized Pauli operators $X(a) = \sum_{j=0}^{q-1} |j+a\rangle \langle j|$, and $Z(a) = \sum_{j=0}^{q-1} \omega^{tr(aj)} |j\rangle \langle j|$, where $q = p^r$ with p being a prime number, $a \in \mathbb{F}_q$ and $\omega = \exp(2\pi i/p)$, then transmit the resulting quantum states to the user over noise-free quantum channels.

In the decoding stage, the user applies a projection-valued measurement (PVM) defined on the quotient space of the stabilizer group $\mathcal{L}(\mathcal{V})$ defined by

$$\mathcal{L}(\mathcal{V}) = \{c_v W(v) : v \in \mathcal{V}\}, \quad (14)$$

where $\mathcal{V}$ is a self-orthogonal subspace, with respect to the symplectic inner product in $\mathbb{F}_q^{2N}$

$$W(v) = X(v_1)Z(v_{N+1}) \otimes \dots \otimes X(v_N)Z(v_{2N}), \quad (15)$$

and $c_v \in \mathbb{C}$ is chosen such that $\mathcal{L}(\mathcal{V})$ is an Abelian subgroup of HW_q^N with $c_v I_{q^N}$ being an element of the stabilizer group only with $c_v = 1$, where HW_q^N is the Heisenberg-Weyl group defined as follows

$$HW_q^N = \{cW(s) : s \in \mathbb{F}_q^{2N}, c \in \mathbb{C} \setminus \{0\}\}. \quad (16)$$

Theorem 4 (Theorem 1 in [37]) Let G be a $2N \times N$ SSO matrix, i.e., $G^t J G = 0$, where $J = \begin{bmatrix} 0 & I_N \\ -I_N & 0 \end{bmatrix}$, and let H be a $2N \times N$ matrix with full column rank. If $[G \ H]$ is a full-rank matrix, then $M = [0_N \ I_N][G \ H]^{-1}$ is a feasible N -sum box transfer matrix.

III. PROBLEM FORMULATION

We now formulate the PDMM problem in the classical and quantum setting, respectively.

In the classical setting, a user possessing two matrices A and B , with constrained local computational resources, seeks to compute their product. These matrices are divided into K and L sub-blocks, respectively, as follows

$$A = \begin{bmatrix} A_1 \\ A_2 \\ \vdots \\ A_K \end{bmatrix}, \quad B = [B_1, B_2, \dots, B_L], \quad (17)$$

where all $A_i B_j$ are compatible for computing the outer product, i.e., AB . Given K and L , the user distributes the encoded versions of the matrices to N servers, such that, these matrices are secure against any T colluding servers, i.e.,

$$I(f_{T_1}, g_{T_2}; A, B) = 0, \quad (18)$$

where f_i and g_i are the information sent by the user to the i th server about A and B , respectively.¹ In addition, $\mathcal{T}_i \subset [N]$, with $|\mathcal{T}_i| \leq T$, $i \in [2]$.

Remark 1 Throughout this paper, we focus on the outer product partitioning case. In this case, we want to compute

$$AB = \begin{bmatrix} A_1 B_1 & A_1 B_2 & \dots & A_1 B_L \\ A_2 B_1 & A_2 B_2 & \dots & A_2 B_L \\ \vdots & \vdots & \ddots & \vdots \\ A_K B_1 & A_K B_2 & \dots & A_K B_L \end{bmatrix}. \quad (19)$$

In order to ensure security for each of these matrices, a subspace with dimension at least T has to be designed to protect the transmitted matrices. For the outer product, AB , this will result in a noise space with high dimensionality. Without careful design, the resulting code may suffer from rate inefficiency.

To address this, we adopt polynomial codes: Polynomial codes are considered as a special case of linear codes due to their nice structure that can be utilized in the analysis. A polynomial code for product computation is defined by two polynomials, $f : \mathbb{F}_q \rightarrow \mathbb{F}_q^{s \times \kappa}$ and $g : \mathbb{F}_q \rightarrow \mathbb{F}_q^{\kappa \times \ell}$, and the goal is to compute $h = fg$ using the N servers. Classically, once the user receives the responses, $h_{[N]}$, the required multiplication can be reconstructed, i.e.,

$$H(AB|h_{[N]}) = 0. \quad (20)$$

In addition, the rate can be defined as

$$R_C(K, L, T) = \frac{KL}{N}. \quad (21)$$

Remark 2 It is important to note that, since only one polynomial, f , is used to encode the sub-blocks of A_i , sub-blocks have to be of equal dimensions, or zero padding must be used. The same argument holds for the sub-blocks of B . In addition, the optimization parameter is the number of servers N , i.e., fewer servers means fewer received bits since the operation is the same for any number of servers.

In the quantum setting, the servers share an entangled state in the quantum system $Q^{\text{init}} = Q_1^{\text{init}} \cdot Q_2^{\text{init}} \cdot \dots \cdot Q_N^{\text{init}}$, where the Q_n^{init} subsystem is at a state represented by the density matrix $\rho_n^{\text{init}} = \text{tr}_{j \in [N] \setminus \{n\}}(\rho^{\text{init}})$, where $\text{tr}_m(\cdot)$ is the partial trace operator in the finite Hilbert space defined in Section II. The user sends the sub-blocks' encoded versions, f_n and g_n , to the n th server over a classical noiseless channel analogous to the classical PDMM protocol. Once the n th server receives the encoded versions, it applies classical operations and encodes over its own quantum state, ρ_n^{init} , to produce another quantum state ρ_n , i.e.,

$$\rho_n = \text{Enc}(\rho_n^{\text{init}}, f_n, g_n). \quad (22)$$

Then, each server transmits ρ_n to the user over a noise-free quantum channel. Upon receiving all quantum subsystems,

¹It is assumed that A and B are generated independently and uniformly at random from the fields $\mathbb{F}_q^{Ks \times \kappa}$, and $\mathbb{F}_q^{\kappa \times \ell L}$, respectively, where $s, \kappa, K, L \in \mathbb{F}_q$.

the user applies an appropriate measurement to recover the classical output, Y , from the quantum states to decode the required multiplication, AB , i.e.,

$$H(AB|Y) = 0. \quad (23)$$

Remark 3 Our encoding and decoding approaches follow the N -sum-box framework described in the subsequent sections.

The efficiency, i.e., rate, for any scheme satisfying (18) and (23) is defined as the ratio between the number of computations and the number of downloaded symbols, i.e.,

$$R(K, L, T) = \frac{mKL}{N}, \quad (24)$$

where m is an integer representing the number of instances for AB that can be computed, i.e., how many multiplications of the form in (19) is obtained in each time the scheme is applied. This is a generalization for the rate formulation; however, in the classical setting, we have $m = 1$ and the classical rate is defined as $R_C(K, L, T)$. In the quantum setting, $m = 2$, which will become clear in later sections. The quantum rate is denoted as $R_Q(K, L, T)$. We note that the number of servers in the quantum setting can potentially be different than the number of servers in the classical setting.

It is important to mention the following theorem that was proven in previous work. The theorem states that, roughly when $T \geq KL$, there is no need to design new quantum codes since the GASP code can be extended to its quantum version. This is useful to know in order to focus our attention to those cases where the design of new quantum codes is needed.

Theorem 5 (Thm. 8 in [38]) If $T \geq KL - K + 1$, with $K, L \geq 2$, then the GASP code can be extended to the quantum counterpart.

IV. A FAMILY OF CODES FOR QUANTUM PDMM COMPUTATIONS IN HIGH PRIVACY REGIME

In this section, we develop new codes for the high privacy regime. These codes do not satisfy the feasibility condition of GASP codes.

A. Main Results

Theorem 6 Let $K = L = T = n^2$, then there exists a quantum scheme based on a polynomial code that achieves a rate equal to

$$R_Q(n^2, n^2, n^2) = \frac{2n^4}{2n^4 + 2n^2 - 1}. \quad (25)$$

Theorem 7 Let $K = L = n^k$, and $T = n^m$, with $m \geq k \geq 2$, then there exists a quantum scheme based on a polynomial code that achieves a rate equal to

$$R_Q(n^k, n^k, n^m) = \frac{2n^{2k}}{2n^{2k} + 3n^m - n^k - 1}. \quad (26)$$

Corollary 1 Let $K = L = n^2$ and $T = n^3$, then there exists a quantum scheme based on a polynomial code that achieves a rate equal to

$$R_Q(n^2, n^2, n^3) = \frac{2n^4}{2n^4 + 3n^3 - n^2 - 1}. \quad (27)$$

Corollary 2 Let $K = L = n^2$ and $T = n^4$, then there exists a quantum scheme based on a polynomial code that achieves a rate equal to

$$R_Q(n^2, n^2, n^4) = \frac{2n^4}{5n^4 - n^2 - 1}. \quad (28)$$

Remark 4 Although Theorem 6 is a special case of Theorem 7, it is presented separately for clarity of comparison. The former is the only closed-form solution for the number of servers in the main GASP paper [23]. Thus, we consider it a stand-alone result.

Theorem 8 Let $K = L = n^k$, and $T = n^k + r$, with $0 \leq r < n^{2k} - n^k + 1$, then there exists a quantum scheme based on a polynomial code that achieves a rate equal to

$$R_Q(n^k, n^k, n^k + r) = \frac{2n^{2k}}{2n^{2k} + 2n^k + 2r - 1}. \quad (29)$$

Theorem 9 For $K \geq L = T$, then there exists a quantum scheme based on the polynomial code that achieves a rate equal to

$$R_Q(K, L, T) = \frac{2KT}{2KT + 2T - 1}. \quad (30)$$

Remark 5 We provide some numerical results and comparisons in order to understand the advantage of the quantum setting over the classical setting when entanglement between servers exists. Let $K = 3$, $L = 2$ and $T = 2$, then the quantum scheme based on a polynomial code achieves a rate equal to

$$R_Q(3, 2, 2) = 0.8, \quad (31)$$

and we have the following gain compared to the classical setting

$$\frac{R_Q(3, 2, 2)}{R_C(3, 2, 2)} = 1.87. \quad (32)$$

Let $K = 4$, $L = T = 2$, then the quantum scheme has a rate equal to

$$R_Q(4, 2, 2) = \frac{16}{19} = 0.84, \quad (33)$$

and in comparison with the classical setting, we have a gain as

$$\frac{R_Q(4, 2, 2)}{R_C(4, 2, 2)} = 1.8. \quad (34)$$

Let $K = 6$, $L = T = 2$, then the quantum scheme achieves a rate equal to

$$R_Q(6, 2, 2) = \frac{24}{27} = 0.89, \quad (35)$$

and the gain compared to the classical scheme is

$$\frac{R_Q(6, 2, 2)}{R_C(6, 2, 2)} = 1.7. \quad (36)$$

Theorem 10 Let $K = T = n^k$ and $L = n^\ell$ with $k \geq \ell$, then there exists a quantum scheme based on a polynomial code that achieves a rate equal to

$$R_Q(n^k, n^\ell, n^k) = \frac{2n^{k+\ell}}{2n^{k+\ell} + 2n^k - 1}. \quad (37)$$

Corollary 3 Let $K = T = 2^3$ and $L = 2$, then

$$\frac{R_Q(8, 2, 8)}{R_C(8, 2, 8)} = 2. \quad (38)$$

Theorem 11 Let $K = T = n^\ell + r$ and $L = n^\ell$ with $r > 0$, then there exists a quantum scheme based on a polynomial code that achieves a rate equal to

$$R_Q(n^\ell + r, n^\ell, n^\ell + r) = \frac{2n^{2\ell} + 2rn^\ell}{2n^{2\ell} + 2rn^\ell + 2n^\ell + 2r - 1}. \quad (39)$$

Remark 6 Note the difference between Theorems 7 and 10 where the first is when $K = L$, and the latter is $K = T$. Similarly, for Theorems 8 and 11.

Remark 7 Although Theorem 11 is a general case of Theorem 10, when substituting $r = n^k - n^\ell$, the number of required servers is more in the latter compared to the former, which decreases the rate. Thus, the designed code for the special case in Theorem 10 is more optimized than the general one in Theorem 11. This is the same for Theorem 7 and Theorem 8.

Remark 8 All the rate comparisons here are with respect to the classical GASP scheme.

As mentioned earlier, Theorem 6 is the only theorem that we can compare analytically with the GASP code since it is the only closed-form result. First, let us recall the GASP code result for the same setting.

Theorem 12 (Proposition 1 in [23]) For $K = L = T = n^2$, the optimal rate for GASP when $n > 1$ is given by

$$R_C(n^2, n^2, n^2) = \frac{n^4}{n^4 + 2n^3 + 2n^2 - n - 2}. \quad (40)$$

Thus, the ratio between the classical and the quantum schemes in this case is given by

$$\frac{R_Q(n^2, n^2, n^2)}{R_C(n^2, n^2, n^2)} = \frac{2n^4 + 4n^3 + 4n^2 - 2n - 4}{2n^4 + 2n^2 - 1}. \quad (41)$$

In comparison with the classical GASP, Figs. 1-5 are plotted to compare for the cases of Theorems 7 and 10. We chose both specifically since one of them has T greater than both K and L , while the other has T equal to K and greater than L . In the first case, Figs. 3, 4, and 5, show that the efficiency of our scheme increases monotonically with T until $T = KL$, with fixed K and L . The efficiency always goes

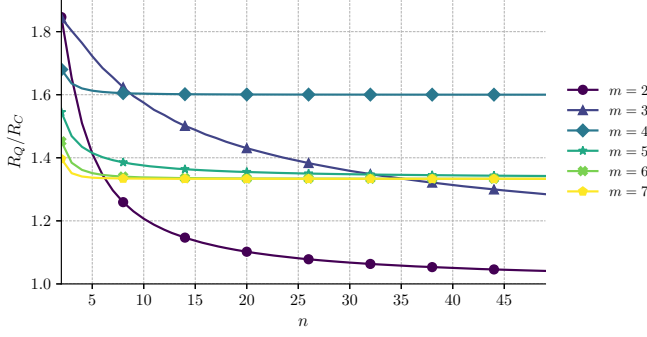


Fig. 1: Comparison between GASP code and the code defined in Theorem 7 when $k = 2$, i.e., $K = L = n^2$ and $T = n^m$, for $m = [2 : 7]$.

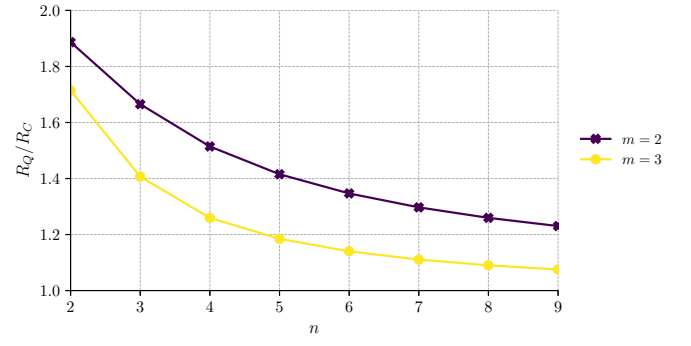


Fig. 3: Comparison between GASP code and the code defined in Theorem 10 when $k = 2$, i.e., $K = T = n^4$ and $L = n^m$, for $m = 2, 3$.

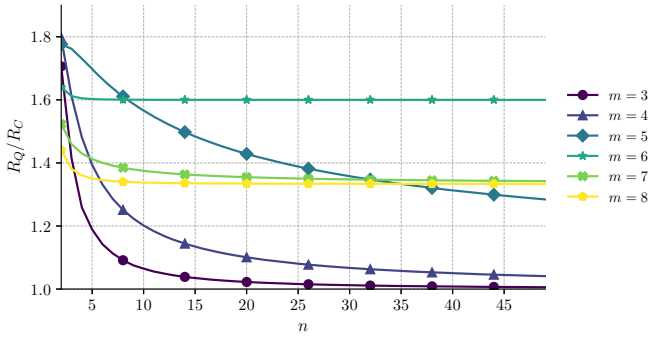


Fig. 2: Comparison between GASP code and the code defined in Theorem 7 when $k = 3$, i.e., $K = L = n^3$ and $T = n^m$, for $m = [3 : 8]$.

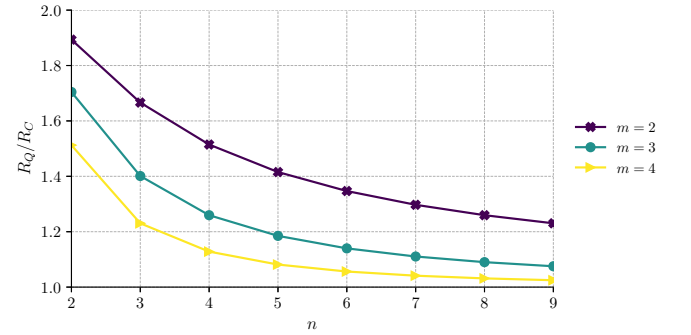


Fig. 4: Comparison between GASP code and the code defined in Theorem 10 when $k = 5$, i.e., $K = T = n^5$ and $L = n^m$, for $m = [2 : 4]$.

to 1 in the case when $T = o(KL)$ since the classical GASP scheme is optimal in the asymptotic sense. However, as T increases, our scheme becomes more efficient than the GASP code even in the asymptotic sense and achieves the highest efficiency when $T = KL$. In addition, in Figs. 3 and 4, the schemes developed can be unnecessary once $m = 2k$ since the feasibility condition is then definitely satisfied, thus achieving the super-dense coding gain. Figs. 3 and 4 show that whenever L increases, the rate gain diminishes for fixed $K = T$.

B. Coding Schemes

In this section, we provide the exponents required for each transmitted polynomial and the encoding of the polynomials for the corresponding code so that the information subspace and the interference subspace become clear for each setting in the previous section. The transfer matrices for all these settings follow the same structure; thus, the transfer matrix and the corresponding encoding are provided only for the first setting. It can be proven that all these matrices satisfy the required properties defined in Theorem 4. In all the following regimes, we assume that the transfer matrices are in the form of

$$M = \begin{bmatrix} 0 & I_N \end{bmatrix} \begin{bmatrix} G & H \end{bmatrix}^{-1}, \quad (42)$$

and it can be shown that the chosen G and H satisfy the required conditions in Theorem 4. In addition, we denote $\alpha =$

(α_1, α_2) and $\beta = (\beta_1, \beta_2)$.

1) $K = L = T = n^2$: The following are the exponents, and the corresponding degree table is given in Table I,

$$\alpha_1 = \beta_1 = [0, 1, \dots, n^2 - 1], \quad (43)$$

$$\alpha_2 = [n^4, n^4 + 1, \dots, n^4 + n^2 - 1], \quad (44)$$

$$\beta_2 = [2n^2 - 1, 3n^2 - 1, \dots, n^4 + n^2 - 1]. \quad (45)$$

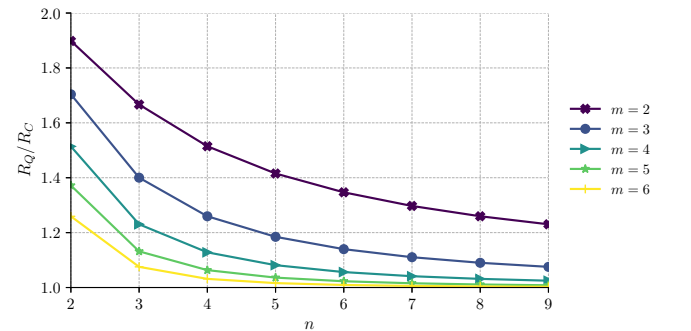


Fig. 5: Comparison between GASP code and the code defined in Theorem 10 when $k = 7$, i.e., $K = T = n^7$ and $L = n^m$, for $m = [2 : 6]$.

$\alpha \backslash \beta$	0	1	$\dots$	$n^2 - 1$	$2n^2 - 1$	$3n^2 - 1$	$\dots$	$n^4 + n^2 - 1$
0	0	1	$\dots$	$n^2 - 1$	$2n^2 - 1$	$3n^2 - 1$	$\dots$	$n^4 + n^2 - 1$
1	1	2	$\dots$	n^2	$2n^2$	$3n^2$	$\dots$	$n^4 + n^2$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
$n^2 - 1$	$n^2 - 1$	n^2	$\dots$	$2n^2 - 2$	$3n^2 - 2$	$4n^2 - 2$	$\dots$	$n^4 + 2n^2 - 2$
n^4	n^4	$n^4 + 1$	$\dots$	$n^4 + n^2 - 1$	$n^4 + 2n^2 - 1$	$n^4 + 3n^2 - 1$	$\dots$	$2n^4 + n^2 - 1$
$n^4 + 1$	$n^4 + 1$	$n^4 + 2$	$\dots$	$n^4 + n^2$	$n^4 + 2n^2$	$n^4 + 3n^2$	$\dots$	$2n^4 + n^2$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
$n^4 + n^2 - 1$	$n^4 + n^2 - 1$	$n^4 + n^2$	$\dots$	$n^4 + 2n^2 - 2$	$n^4 + 3n^2 - 2$	$n^4 + 4n^2 - 2$	$\dots$	$2n^4 + 2n^2 - 2$

TABLE I: Degree table for $K = L = T = n^2$.

The encoding is done as follows,

$$f(x) = \sum_{i=1}^{n^2} R_i x^{\alpha(i)} + \sum_{i=1}^{n^2} A_i x^{\alpha(i+n^2)}, \quad (46)$$

$$g(x) = \sum_{i=1}^{n^2} Z_i x^{\beta(i)} + \sum_{i=1}^{n^2} B_i x^{\beta(i+n^2)}. \quad (47)$$

As an illustrative example, we provide a detailed derivation of the transmitted and received symbols along with the chosen transfer matrix. The remaining schemes follow the same structure. The user sends the two instances of the aforementioned polynomials to compute $A^{[1]}B^{[1]}$ and $A^{[2]}B^{[2]}$, where each resembles the structure in (19), i.e., for $m \in [2]$,

$$f^{[m]}(a_i) = \sum_{t=1}^{n^2} R_k^{[m]} a_i^{\alpha(t)} + \sum_{k=1}^{n^2} A_k^{[m]} a_i^{\alpha(k+n^2)}, \quad (48)$$

$$g^{[m]}(a_i) = \sum_{t=1}^{n^2} Z_t^{[m]} a_i^{\beta(t)} + \sum_{k=1}^{n^2} B_k^{[m]} a_i^{\beta(k+n^2)}, \quad (49)$$

to the i th server. Upon receiving them, each server applies the multiplication and computes

$$F^{[m]}(a_n) = \sum_{(k,\ell) \in \mathcal{Q}} A_k^{[m]} B_\ell^{[m]} a_n^{(\alpha \oplus \beta)_{k,\ell}} + \sum_{(i,j) \in \mathcal{Q}} I_{i,j}^{[m]} a_n^{(\alpha \oplus \beta)_{i,j}}, \quad m \in [2], \quad (50)$$

where $\mathcal{Q}$ is the interference subspace with $i \in [K]$, or $j \in [L]$. Now, for each element in $F_n^{[1]} = F^{[1]}(a_n)$, $F_n^{[2]} = F^{[2]}(a_n)$, the server applies Pauli operators on each element as follows

$$\begin{aligned} & \mathbf{X}(u_n F_n^{[1]}(i, j)) \mathbf{Z}(v_n F_n^{[2]}(i, j)) \rho_n \\ & \times \mathbf{X}^\dagger(u_n F_n^{[1]}(i, j)) \mathbf{Z}^\dagger(v_n F_n^{[2]}(i, j)). \end{aligned} \quad (51)$$

The transmitted symbols can be written as follows,

$$\begin{aligned} X(i, j) &= [F_1^{[1]}(i, j), \dots, F_{2n^4+2n^2-1}^{[1]}(i, j), \\ & F_1^{[2]}(i, j), \dots, F_{2n^4+2n^2-1}^{[2]}(i, j)]^t \\ &= \begin{bmatrix} \text{diag}(u)G' & 0_{2n^4+2n^2-1} \\ 0_{2n^4+2n^2-1} & \text{diag}(v)G' \end{bmatrix} \end{aligned} \quad (52)$$

$$\begin{aligned} & \times \left[\underbrace{I_1^{[1]}(i, j), \dots, I_{n^4+2n^2-1}^{[1]}(i, j)}_{\text{Interference space 1}}, (A_1 B_1)^{[1]}(i, j), \right. \\ & (A_1 B_2)^{[1]}(i, j), \dots, (A_K B_L)^{[1]}(i, j), \\ & \underbrace{I_1^{[2]}(i, j), \dots, I_{n^4+2n^2-1}^{[2]}(i, j)}_{\text{Interference space 2}}, (A_1 B_1)^{[2]}(i, j), \\ & \left. \dots, (A_K B_L)^{[2]}(i, j) \right]^t, \end{aligned} \quad (53)$$

where

$$G' = \begin{bmatrix} 1 & a_1 & a_1^2 & \dots & a_1^{2n^4+2n^2-2} \\ 1 & a_2 & a_2^2 & \dots & a_2^{2n^4+2n^2-2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & a_{2n^4+2n^2-1} & a_{2n^4+2n^2-1}^2 & \dots & a_{2n^4+2n^2-1}^{2n^4+2n^2-2} \end{bmatrix}. \quad (54)$$

There are two interference vector spaces as shown in (52), each for a separate classical instance of the code, i.e., two multiplications, $AB^{[1]}$ and $AB^{[2]}$. The transfer matrix M for the N -sum box should be chosen such that the maximum number of symbols in the first and second interference vector spaces are dropped. This must be done while using a valid transfer matrix M , i.e., full rank and satisfying the SSO condition. Thus, M is chosen as $M = [0 \ I_N] [G \ H]^{-1}$, with

$$G = D \begin{bmatrix} G'_1 & 0 \\ 0 & G'_2 \end{bmatrix}, \quad (55)$$

with $G'_1 = G'(:, 1 : \lfloor \frac{2n^4+2n^2-1}{2} \rfloor - 1)$, $G'_2 = G'(:, 1 : \lceil \frac{2n^4+2n^2-1}{2} \rceil - 1)$ and

$$H = D \begin{bmatrix} H_1 & 0 \\ 0 & H_2 \end{bmatrix}, \quad (56)$$

with $H_1 = G'(:, \lfloor \frac{2n^4+2n^2-1}{2} \rfloor : \text{end})$, $H_2 = G'(:, \lceil \frac{2n^4+2n^2-1}{2} \rceil : \text{end})$, and

$$D = \begin{bmatrix} \text{diag}(u) & 0 \\ 0 & \text{diag}(v) \end{bmatrix}. \quad (57)$$

Thus, the symbols that the user gets after measurement can be

written as,

$$\begin{aligned}
 Y(i, j) &= MX(i, j) \\
 &= \underbrace{\left[I_{\lfloor \frac{2n^4+2n^2-1}{2} \rfloor}^{[1]}(i, j), \dots, I_{n^4+2n^2-1}^{[1]}(i, j), \right.}_{\text{Remaining interference subspace 1}} \\
 &\quad (A_1 B_1)^{[1]}(i, j), \dots, (A_{n^2} B_{n^2})^{[1]}(i, j), \\
 &\quad \underbrace{\left. I_{\lceil \frac{2n^4+2n^2-1}{2} \rceil}^{[2]}(i, j), \dots, I_{n^4+2n^2-1}^{[2]}(i, j), \right.}_{\text{Remaining interference subspace 2}} \\
 &\quad \left. (A_1 B_1)^{[2]}(i, j), \dots, (A_{n^2} B_{n^2})^{[2]}(i, j) \right]^t, \quad (59)
 \end{aligned}$$

where as shown in (58), the remaining interference subspace has lower dimensionality compared to the one in (52). The first interference subspace in (58) starts from $\lfloor \frac{2n^4+2n^2-1}{2} \rfloor$ as opposed to starting at 1 in (52). Similarly, the second interference subspace in (58) starts from $\lceil \frac{2n^4+2n^2-1}{2} \rceil$. Thus, from (58), upon measurement, the user receives two instances of AB , i.e., $AB^{[1]}$, and $AB^{[2]}$, over a subspace of dimension $2n^4$.

2) $K = L = n^k$ and $T = n^m$: The following are the exponents, and the corresponding degree table is given in Table II,

$$\alpha_1 = \beta_1 = [0, 1, \dots, n^m - 1], \quad (60)$$

$$\alpha_2 = [n^m + n^{2k} - n^k, n^m + n^{2k} - n^k + 1, \dots, n^m + n^{2k} - 1], \quad (61)$$

$$\beta_2 = [2n^m - 1, 2n^m + n^k - 1, \dots, 2n^m + n^{2k} - n^k - 1]. \quad (62)$$

The encoding is done as follows,

$$f(x) = \sum_{i=1}^{n^m} R_i x^{\alpha(i)} + \sum_{i=1}^{n^k} A_i x^{\alpha(i+n^m)}, \quad (63)$$

$$g(x) = \sum_{i=1}^{n^m} Z_i x^{\beta(i)} + \sum_{i=1}^{n^k} B_i x^{\beta(i+n^m)}. \quad (64)$$

3) $K = L = n^k$ and $T = n^k + r$: The following are the exponents, and the corresponding degree table is given in Table III,

$$\alpha_1 = \beta_1 = [0, 1, \dots, n^k + r - 1], \quad (65)$$

$$\alpha_2 = [n^{2k}, n^{2k} + 1, \dots, n^{2k} + n^k - 1], \quad (66)$$

$$\beta_2 = [2n^k + 2r - 1, 3n^k + 2r - 1, \dots, n^{2k} + n^k + 2r - 1]. \quad (67)$$

The encoding is done as follows,

$$f(x) = \sum_{i=1}^T R_i x^{\alpha(i)} + \sum_{i=1}^K A_i x^{\alpha(i+T)}, \quad (68)$$

$$g(x) = \sum_{i=1}^T Z_i x^{\beta(i)} + \sum_{i=1}^L B_i x^{\beta(i+T)}. \quad (69)$$

4) $K \geq L$ and $L = T$: The following are the exponents and the corresponding degree table is given in Table IV,

$$\alpha_1 = \beta_1 = [0, 1, \dots, T - 1], \quad (70)$$

$$\alpha_2 = [2T - 1, 3T - 1, \dots, KT + T - 1], \quad (71)$$

$$\beta_2 = [KT, KT + 1, \dots, KT + T - 1]. \quad (72)$$

The encoding is done as follows

$$f(x) = \sum_{i=1}^T R_i x^{\alpha(i)} + \sum_{i=1}^K A_i x^{\alpha(T+i)}, \quad (73)$$

$$g(x) = \sum_{i=1}^T Z_i x^{\beta(i)} + \sum_{i=1}^L B_i x^{\beta(T+i)}. \quad (74)$$

5) $K = T = n^k$ and $L = n^\ell$: The following are the exponents, and the corresponding degree table is given in Table V,

$$\alpha_1 = \beta_1 = [0, 1, \dots, n^k - 1], \quad (75)$$

$$\alpha_2 = [n^{\ell+k}, n^{\ell+k} + 1, \dots, n^{\ell+k} + n^k - 1], \quad (76)$$

$$\beta_2 = [2n^k - 1, 3n^k - 1, \dots, n^{k+\ell} + n^k - 1]. \quad (77)$$

The encoding is done as follows,

$$f(x) = \sum_{i=1}^T R_i x^{\alpha(i)} + \sum_{i=1}^K A_i x^{\alpha(T+i)}, \quad (78)$$

$$g(x) = \sum_{i=1}^T Z_i x^{\beta(i)} + \sum_{i=1}^L B_i x^{\beta(i+T)}. \quad (79)$$

6) $K = T = n^\ell + r$ and $L = n^\ell$: The following are the exponents, and the corresponding degree table is given in Table VI.

$$\alpha_1 = \beta_1 = [0, 1, \dots, n^\ell + r - 1], \quad (80)$$

$$\alpha_2 = [2n^\ell + r - 1, 3n^\ell + r - 1, \dots, n^{2\ell} + (r+1)n^\ell + r - 1], \quad (81)$$

$$\beta_2 = [n^{2\ell} + rn^\ell + r, n^{2\ell} + rn^\ell + r + 1, \dots, n^{2\ell} + (r+1)n^\ell + r - 1]. \quad (82)$$

The encoding is done as follows,

$$f(x) = \sum_{i=1}^T R_i x^{\alpha(i)} + \sum_{i=1}^K A_i x^{\alpha(T+i)}, \quad (83)$$

$$g(x) = \sum_{i=1}^T Z_i x^{\beta(i)} + \sum_{i=1}^L B_i x^{\beta(i+T)}. \quad (84)$$

C. Discussion and Further Remarks

First, we note that, although in some cases GASP codes are not extendable to their quantum counterparts, our code has the same number of servers as GASP, thus achieving super-dense coding gain as in the case with $K = T = 8$ and $L = 2$.

Second, it is important to note that when GASP is asymptotically optimal, like the case when $K = L = T = n^2$, the asymptotic ratio between the classical and quantum schemes is 1 due to the Holevo bound. To state the bound simply, in a classical-quantum setting, where the entangled transmitters (in our setting, N entangled transmitters) have classical information to encode over quantum states, N bits of information can be received by the receiver. Consequently, as $n \rightarrow \infty$, we have $KL \rightarrow N$ as shown in Theorem 12. Thus, the highest possible

$\alpha \backslash \beta$	0	1	$\dots$	$n^m - 1$	$2n^m - 1$	$2n^m + n^k - 1$	$\dots$	$2n^m + n^{2k} - n^k - 1$
0	0	$\dots$	$\dots$	$n^m - 1$	$2n^m - 1$	$\dots$	$\dots$	$2n^m + n^{2k} - n^k - 1$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
$n^m - 1$	$n^m - 1$	$\dots$	$\dots$	$2n^m - 2$	$3n^m - 2$	$\dots$	$\dots$	$3n^m + n^{2k} - n^k - 2$
$n^m + n^{2k} - n^k$	$n^m + n^{2k} - n^k$	$\dots$	$\dots$	$2n^m + n^{2k} - n^k - 1$	$3n^m + n^{2k} - n^k - 1$	$\dots$	$\dots$	$3n^m + 2n^{2k} - 2n^k - 1$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
$n^m + n^{2k} - 1$	$n^m + n^{2k} - 1$	$\dots$	$\dots$	$2n^m + n^{2k} - 2$	$3n^m + n^{2k} - 2$	$\dots$	$\dots$	$3n^m + 2n^{2k} - n^k - 2$

TABLE II: Degree table for $K = L = n^k$, and $T = n^m$.

$\alpha \backslash \beta$	0	1	$\dots$	$n^k + r - 1$	$2n^k + 2r - 1$	$3n^k + 2r - 1$	$\dots$	$n^{2k} + n^k + 2r - 1$
0	0	$\dots$	$\dots$	$n^k + r - 1$	$2n^k + 2r - 1$	$\dots$	$\dots$	$n^{2k} + n^k + 2r - 1$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
$n^k + r - 1$	$n^k + r - 1$	$\dots$	$\dots$	$2n^k + 2r - 2$	$3n^k + 3r - 2$	$\dots$	$\dots$	$n^{2k} + 2n^k + 2r - 2$
n^{2k}	n^{2k}	$\dots$	$\dots$	$n^{2k} + n^k + r - 1$	$n^{2k} + 2n^k + 2r - 1$	$\dots$	$\dots$	$2n^{2k} + n^k + 2r - 1$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
$n^{2k} + n^k - 1$	$n^{2k} + n^k - 1$	$\dots$	$\dots$	$n^{2k} + 2n^k + r - 2$	$n^{2k} + 3n^k + 2r - 1$	$\dots$	$\dots$	$2n^{2k} + 2n^k + 2r - 2$

TABLE III: Degree table for $K = L = n^k$, and $T = n^m$.

$\alpha \backslash \beta$	0	1	$\dots$	$T - 1$	KT	$KT + 1$	$\dots$	$KT + T - 1$
0	0	$\dots$	$\dots$	$T - 1$	KT	$\dots$	$\dots$	$KT + T - 1$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
$T - 1$	$T - 1$	$\dots$	$\dots$	$2T - 2$	$KT + T - 1$	$\dots$	$\dots$	$KT + 2T - 2$
$2T - 1$	$2T - 1$	$\dots$	$\dots$	$3T - 2$	$KT + 2T - 1$	$\dots$	$\dots$	$KT + 3T - 2$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
$KT + T - 1$	$KT + T - 1$	$\dots$	$\dots$	$KT + 2T - 2$	$2KT + T - 1$	$\dots$	$\dots$	$2KT + 2T - 2$

TABLE IV: Degree table for $K \geq L = T$.

asymptotic quantum rate is also 1 for entangled transmitters in the quantum PDMM setting.

V. A FAMILY OF CODES FOR PDMM IN THE LOW PRIVACY REGIME

Although many regimes are covered in the previous sections, the case where $T < \min(K, L) = L$ is still the main missing piece in our work. An interesting approach to look at is to increase the security parameters to one of the polynomials, i.e., instead of

$$g(x) = \sum_{i=1}^L B_i x^{\beta(i)} + \sum_{j=1}^T R_j x^{\beta(j+L)}, \quad (85)$$

we can extend this to $\bar{T} > T$, such that

$$g(x) = \sum_{i=1}^L B_i x^{\beta(i)} + \sum_{j=1}^{\bar{T}} R_j x^{\beta(j+L)}. \quad (86)$$

The $\bar{T}$ is added in a way that makes the transfer matrix extendable to the quantum version. As a starting example, let $K = L = 2$ and $T = 1$. As shown in [22], the GASP exponents are given by

$$\alpha = [0, 2, 4], \quad \beta = [0, 1, 4]. \quad (87)$$

The rate in this case is given by $R_C = \frac{1}{2}$. Unfortunately, this is not directly extendable to the quantum version. However, if we change the code to the following exponents

$$\alpha = [0, 2, 4], \quad \beta = [0, 1, 4, 5], \quad (88)$$

then this can be extended to the quantum version using the scheme in [34], and the rate is given by $R_Q = \frac{4}{5}$.

Another example is when $K = L = 3$ and $T = 1$. In this case, we have the following exponents for the GASP

$$\alpha = [0, 1, 2, 9], \quad \beta = [0, 3, 6, 9], \quad (89)$$

with the information encoded over the first three elements of each exponent vector, α and β . The rate in this case is

$\alpha \backslash \beta$	0	1	$\dots$	$n^k - 1$	$2n^k - 1$	$3n^k - 1$	$\dots$	$n^{k+\ell} + n^k - 1$
0	0	$\dots$	$\dots$	$n^k - 1$	$2n^k - 1$	$\dots$	$\dots$	$n^{k+\ell} + n^k - 1$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
$n^k - 1$	$n^k - 1$	$\dots$	$\dots$	$2n^k - 2$	$3n^k - 2$	$\dots$	$\dots$	$n^{k+\ell} + 2n^k - 2$
$n^{k+\ell}$	$n^{k+\ell}$	$\dots$	$\dots$	$n^{k+\ell} + n^k - 1$	$n^{k+\ell} + 2n^k - 1$	$\dots$	$\dots$	$2n^{k+\ell} + n^k - 1$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
$n^{k+\ell} + n^k - 1$	$n^{k+\ell} + n^k - 1$	$\dots$	$\dots$	$n^{k+\ell} + 2n^k - 2$	$n^{k+\ell} + 3n^k - 2$	$\dots$	$\dots$	$2n^{k+\ell} + 2n^k - 2$

TABLE V: Degree table for $K = T = n^k$ and $L = n^\ell$.

$\alpha \backslash \beta$	0	1	$\dots$	$n^\ell + r - 1$	$n^{2\ell} + rn^\ell + r$	$n^{2\ell} + rn^\ell + r + 1$	$\dots$	$n^{2\ell} + (r+1)n^\ell + r - 1$
0	0	$\dots$	$\dots$	$n^\ell + r - 1$	$n^{2\ell} + rn^\ell + r$	$\dots$	$\dots$	$n^{2\ell} + (r+1)n^\ell + r - 1$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
$n^\ell + r - 1$	$n^\ell + r - 1$	$\dots$	$\dots$	$2n^\ell + 2r - 2$	$n^{2\ell} + (r+1)n^\ell + 2r - 1$	$\dots$	$\dots$	$n^{2\ell} + (r+2)n^\ell + 2r - 2$
$2n^\ell + r - 1$	$2n^\ell + r - 1$	$\dots$	$\dots$	$3n^\ell + 2r - 2$	$n^{2\ell} + (r+2)n^\ell + 2r - 1$	$\dots$	$\dots$	$n^{2\ell} + (r+3)n^\ell + 2r - 2$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
$n^{2\ell} + (r+1)n^\ell + r - 1$	$n^{2\ell} + (r+1)n^\ell + r - 1$	$\dots$	$\dots$	$n^{2\ell} + (r+2)n^\ell + 2r - 2$	$2n^{2\ell} + (2r+1)n^\ell + 2r - 1$	$\dots$	$\dots$	$2n^{2\ell} + (2r+2)n^\ell + 2r - 2$

TABLE VI: Degree table for $K = T = n^\ell + r$, and $L = n^\ell$.

$R_C = \frac{9}{15}$. However, this is very hard to change directly with the information sub-space being the same like the previous example. If we choose $\bar{T} = 3$, we can design the following exponents

$$\alpha = [0, 3, 5, 7], \quad \beta = [0, 1, 2, 10, 18, 26], \quad (90)$$

achieving a rate equal to $R_Q = \frac{18}{22} = \frac{9}{11}$. Thus, the approach that we take is to use $\bar{T}$ instead of T in one of the polynomials and encode the useful information over the higher degree part of the resulting polynomial.

A. Main Results

Theorem 13 Let $K \geq L \geq T$, with $2mL \geq \delta L + (m-2)(T-1) + (m-1)L^2 + 6$. Then, there exists a quantum code that is based on a polynomial code that satisfies the PDMM requirements, achieving a rate equal to

$$R_Q = \frac{2KL}{K(L+3) + (m+2)(T-1) + L^2 - 2L}, \quad (91)$$

where $K = m(L-1) + \delta$ and $0 \leq \delta < L-1$.

Corollary 4 If $K = L > T$ with $L + T \geq 7$, then there exists a quantum code that is based on a polynomial code that satisfies the PDMM requirements, achieving a rate equal to

$$R_Q = \frac{2L^2}{2L^2 + L + 3T - 3}. \quad (92)$$

To understand the ratio between the highest classical rate and our achieved rate in the low privacy regime, Figs. 6-9 are shown for numerical representation for the $K = L > T$

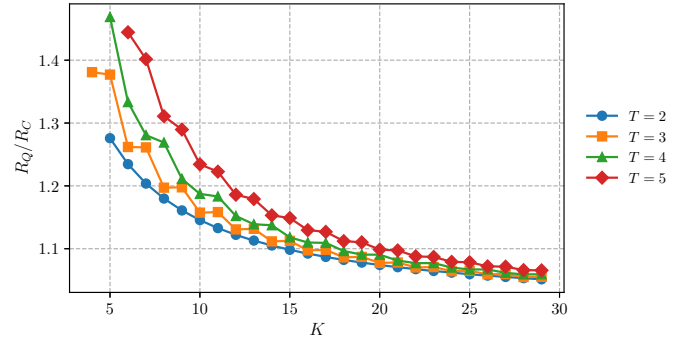


Fig. 6: Comparison between the highest rate classical code and the code defined in Corollary 4 when $K = L$ and $T = [2 : 5]$.

case since this case is less complicated. The figures show two things, the first is that when T is small, the code designed is not going to be as efficient as in the high privacy regimes. This is mainly because the interference subspace, i.e., the subspace that carries no information about the multiplication AB , is small compared to the whole space, which makes it harder to optimize since fewer terms can be dropped. The second reason is that the classical codes are efficient so that the rate is close enough to unity, which makes the quantum codes not being able to surpass the classical counterparts by much due to the Holevo bound.

B. Encoding Structure

To make it easier to visualize, we first start by proving Corollary 4, and then go to the more general setting. The

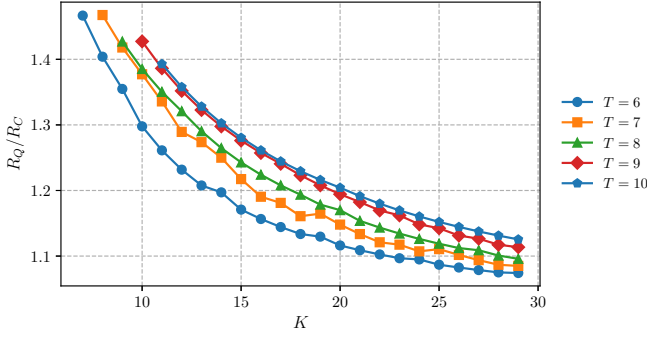


Fig. 7: Comparison between the highest rate classical code and the code defined in Corollary 4 when $K = L$ and $T = [6 : 10]$.

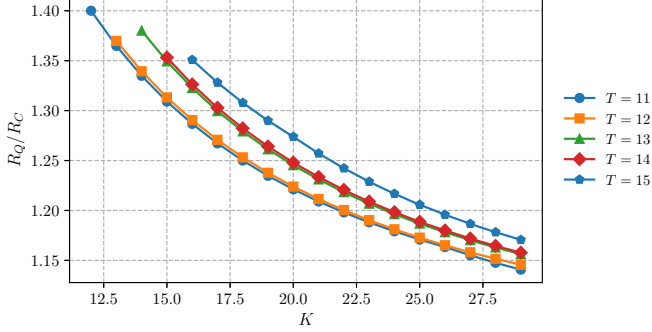


Fig. 8: Comparison between the highest rate classical code and the code defined in Corollary 4 when $K = L$ and $T = [11 : 15]$.

construction of the G and H matrices follows similar to Section IV-B.

1) $K = L > T$: The following are the exponents, and the corresponding degree table is given in Table VII,

$$\alpha_1 = [0 : L - 1], \quad (93)$$

$$\beta_1 = [0 : T - 1], \quad (94)$$

$$\alpha_2 = \underbrace{[L^2 + T - 1, L^2 + T, \dots, L^2 + L + T - 3, 2L^2 + 2T - 3]}_{L-1}, \quad (95)$$

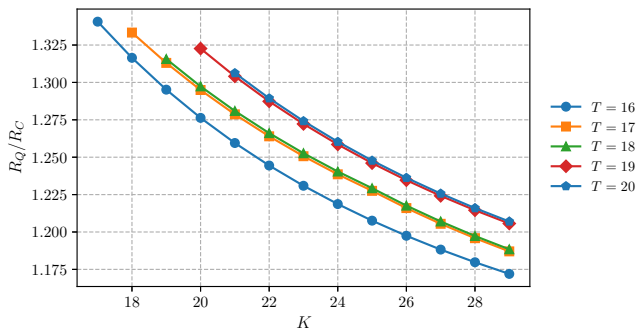


Fig. 9: Comparison between the highest rate classical code and the code defined in Corollary 4 when $K = L$ and $T = [16 : 20]$.

$$\beta_2 = [T + L - 2, T + 2L - 3, \dots, L^2 - L + T - 1]. \quad (96)$$

The encoding is done as follows,

$$f(x) = \sum_{i=1}^L R_i x^{\alpha(i)} + \sum_{j=1}^L A_j x^{\alpha(T+j)}, \quad (97)$$

$$g(x) = \sum_{i=1}^T Z_i x^{\beta(i)} + \sum_{j=1}^L B_j x^{\beta(T+j)}. \quad (98)$$

To show that the code is a valid quantum PDMM code, since there are gaps in the last row, we check the longest chain of consecutive exponents in the degree table for the interference subspace, i.e., the part of the degree Table VII excluding $\alpha_2 \oplus \beta_2$. Thus, the longest chain is $[0 : L^2 + L + 2T - 4]$. To that end, we need to make sure that it is greater than the ceiling of half of the number of servers, i.e., $L^2 + L + 2T - 4 \geq \lceil \frac{2L^2 + L + 3T - 3}{2} \rceil$. To proceed, we need to prove that $L^2 + L + 2T - 4 \geq \lceil \frac{2L^2 + L + 3T - 3}{2} \rceil$. Recall that $L + T \geq 7$, then we have

$$\frac{L + T}{2} \geq 3.5 \Rightarrow L^2 + L + 2T - 4 \geq L^2 + \frac{L}{2} + 1.5T - 0.5 \quad (99)$$

$$\Rightarrow L^2 + L + 2T - 4 \geq \frac{2L^2 + L + 3T - 3}{2} + 1 \quad (100)$$

$$\Rightarrow L^2 + L + 2T - 4 \geq \left\lceil \frac{2L^2 + L + 3T - 3}{2} \right\rceil. \quad (101)$$

2) $K \geq L > T$: First, assume that $K = m(L - 1) + \delta$, $0 \leq \delta < L - 1$. Then, the following are the exponents and the corresponding degree table is given in Table VIII,

$$\alpha_1 = [0 : K - 1], \quad (102)$$

$$\beta_1 = [0 : T - 1], \quad (103)$$

$$\alpha'_1 = [2K + T + L^2 - 2L - 1, 2K + T + L^2 - 2L, \dots, 2K + T + L^2 - L - 3], \quad (104)$$

$$\alpha'_2 = [3K + 2T + 2L^2 - 2L - 3, 3K + 2T + 2L^2 - 2L - 2, \dots, 3K + 2T + 2L^2 - 2L - 5], \quad (105)$$

$\vdots$

$$\begin{aligned} \alpha'_m = & [(m+1)K + mT + mL^2 - (m+1)L - 2m + 1, \\ & (m+1)K + mT + mL^2 - (m+1)L - 2m + 2, \\ & \dots, (m+1)K + mT + mL^2 - mL - 2m - 1], \end{aligned} \quad (106)$$

$$\begin{aligned} \alpha'' = & [(m+2)K + (m+1)T + (m+1)L^2 - (m+2)L \\ & - 2m - 1, (m+2)K + (m+1)T + (m+1)L^2 \\ & - (m+2)L - 2m, \dots, (m+2)K + (m+1)T \\ & + (m+1)L^2 - (m+2)L - 2m + \delta - 2] \end{aligned} \quad (107)$$

$$\beta_2 = [K + T - 2, K + T + L - 3, \dots, K + T + L^2 - 2L - 1], \quad (108)$$

where $\alpha_2 = [\alpha'_1, \dots, \alpha'_m, \alpha'']$, with each α'_i , $i \in [m]$ is of length $L - 1$, and α'' is of length δ .

$\alpha \backslash \beta$	0	1	$\dots$	$T-1$	$T+L-2$	$T+2L-3$	$\dots$	$L^2-L+T-1$
0	0	$\dots$	$\dots$	$T-1$	$T+L-2$	$\dots$	$\dots$	$L^2-L+T-1$
1	$\vdots$	$\ddots$	$\ddots$	$\vdots$	$\vdots$	$\ddots$	$\ddots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
$L-1$	$L-1$	$\dots$	$\dots$	$T+L-2$	$T+2L-3$	$\dots$	$\dots$	L^2+T-2
L^2+T-1	L^2+T-1	$\dots$	$\dots$	L^2+2T-2	$L^2+L+2T-3$	$\dots$	$\dots$	$2L^2-L+2T-2$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
$L^2+L+T-3$	$L^2+L+T-3$	$\dots$	$\dots$	$L^2+L+2T-4$	$L^2+2L+2T-5$	$\dots$	$\dots$	$2L^2+2T-4$
$2L^2+2T-3$	$2L^2+2T-3$	$\dots$	$\dots$	$2L^2+3T-4$	$2L^2+3T+L-5$	$2L^2+3T+2L-6$	$\dots$	$3L^2-L+3T-4$

TABLE VII: Degree table for $K = L > T$.

$\alpha \backslash \beta$	0	1	$\dots$	$T-1$	$K+T-2$	$K+T+L-3$	$\dots$	$K+T+L^2-2L-1$
0	0	$\dots$	$\dots$	$T-1$	$K+T-2$	$\dots$	$\dots$	$K+T+L^2-2L-1$
1	$\vdots$	$\ddots$	$\ddots$	$\vdots$	$\vdots$	$\ddots$	$\ddots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
$K-1$	$K-1$	$\dots$	$\dots$	$K+T-2$	$2K+T-3$	$\dots$	$\dots$	$2K+T+L^2-2L-2$
$2K+T+L^2-2L-1$	$2K+T+L^2-2L-1$	$\dots$	$\dots$	$2K+2T+L^2-2L-2$	$3K+2T+L^2-2L-3$	$\dots$	$\dots$	$3K+2T+2L^2-4L-2$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
$2K+T+L^2-L-3$	$2K+T+L^2-L-3$	$\dots$	$\dots$	$2K+2T+L^2-L-4$	$3K+2T+L^2-L-5$	$\dots$	$\dots$	$3K+2T+2L^2-3L-4$
$3K+2T+2L^2-2L-3$	$3K+2T+2L^2-2L-3$	$\dots$	$\dots$	$3K+3T+2L^2-2L-4$	$4K+3T+2L^2-2L-5$	$\dots$	$\dots$	$4K+3T+3L^2-4L-4$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
$3K+2T+2L^2-L-5$	$3K+2T+2L^2-L-5$	$\dots$	$\dots$	$3K+3T+2L^2-2L-6$	$4K+3T+2L^2-2L-7$	$\dots$	$\dots$	$4K+3T+3L^2-3L-6$
$\vdots$	$\dots$	$\dots$	$\dots$	$\vdots$	$\dots$	$\dots$	$\dots$	$\dots$
$(m+1)K+mT+mL^2-(m+1)L-2m+1$	$(m+1)K+mT+mL^2-(m+1)L-2m+1$	$\dots$	$\dots$	$(m+1)K+(m+1)T+mL^2-(m+1)L-2m$	$(m+2)K+(m+1)T+mL^2-(m+1)L-2m-1$	$\dots$	$\dots$	$(m+2)K+(m+1)T+(m+1)L^2-(m+3)L-2m$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
$(m+1)K+mT+mL^2-mL-2m-1$	$(m+1)K+mT+mL^2-mL-2m-1$	$\dots$	$\dots$	$(m+1)K+(m+1)T+mL^2-mL-2m$	$(m+2)K+(m+1)T+mL^2-mL-2m-3$	$\dots$	$\dots$	$(m+2)K+(m+1)T+(m+1)L^2-(m+2)L-2m-2$
$(m+2)K+(m+1)T+(m+1)L^2-(m+2)L-2m-1$	$(m+2)K+(m+1)T+(m+1)L^2-(m+2)L-2m-1$	$\dots$	$\dots$	$(m+2)K+(m+2)T+(m+1)L^2-(m+2)L-2m-2$	$(m+3)K+(m+2)T+(m+1)L^2-(m+2)L-2m-3$	$\dots$	$\dots$	$(m+3)K+(m+2)T+(m+2)L^2-(m+4)L-2m-1$
$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$	$\vdots$	$\dots$	$\vdots$
$(m+2)K+(m+1)T+(m+1)L^2-(m+2)L-2m+\delta-2$	$(m+2)K+(m+1)T+(m+1)L^2-(m+2)L-2m+\delta-2$	$\dots$	$\dots$	$(m+2)K+(m+2)T+(m+1)L^2-(m+2)L-2m+\delta-3$	$(m+3)K+(m+2)T+(m+1)L^2-(m+2)L-2m+\delta-4$	$\dots$	$\dots$	$(m+3)K+(m+2)T+(m+2)L^2-(m+4)L-2m+\delta-3$

TABLE VIII: Degree table for $K \geq L > T$.

The encoding is done as follows

$$f(x) = \sum_{i=1}^K R_i x^{\alpha(i)} + \sum_{j=1}^K A_j x^{\alpha(K+j)}, \quad (109)$$

$$g(x) = \sum_{i=1}^T Z_i x^{\beta(i)} + \sum_{j=1}^L B_j x^{\beta(T+j)}. \quad (110)$$

C. Discussion and Further Remarks

In this section, we proposed a method to address the low privacy regime case, where T is small compared to K and L . There might be room for refinement in this case, as suggested by observations in the classical setting, where different codes compete for efficiency in this regime with no single scheme demonstrating consistent superiority. We proposed extending only one polynomial with an extra privacy requirement so that

it becomes extendable to the quantum version. As shown in the figures presented in this section, the gain is at most 1.5. This suggests that more efficient code constructions may exist.

A particularly promising yet challenging direction is the entanglement-assisted framework, reformulated recently in [33], where local precoding matrix is used at each server. The main obstacle is to find the precoding that can enhance the achievable rate.

VI. CONCLUSION

In this paper, we investigated the problem of private distributed matrix multiplication (PDMM) for the case of OPP in the quantum setting, i.e., the QPDMM problem. First, we noted that the classical codes for the PDMM problem are not always extendable to their quantum counterpart. Second, we divided our approach into two regimes based on the privacy requirement. In the high privacy regime, we designed several quantum schemes based on the system parameters in order to compensate for the inapplicability of the extension of the classical codes. We compared the designed quantum codes and their counterpart of the state-of-the-art classical code in several settings. We showed that super-dense coding gain can be achieved in some cases. In addition, we observed that the gain plots show that the gain ranges from 1.8 to 1 in this regime. We then designed quantum codes for the low privacy regime. The design for these codes is based on previous work on the quantum version of the private information retrieval (QPIR) problem by extending the privacy requirement slightly. The plots show that the gain varies from 1.5 to 1 in this regime.

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